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POSTER ABSTRACTS

803.EMERGING TOOLS, TECHNIQUES AND ARTIFICIAL INTELLIGENCE IN HEMATOLOGY

Diagnostic Performance and Clinical Impacts of Metagenomic Sequencing after Allogeneic Hematopoietic Stem Cell Transplantation

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Background: Metagenomic Next-Generation Sequencing (mNGS) is a rapid, non-culture-based, and high-throughput technique for pathogen diagnosis. Despite its numerous advantages, only a few studies have investigated its use in patients undergoing allogeneic hematopoietic stem cell transplantation (allo-HSCT).

Methods: We conducted a retrospective analysis of 404 mNGS tests (343 infectious episodes) performed on 264 patients following allo-HSCT. We assessed the analytical performance and clinical utility of mNGS for clinical impacts.

Results: Metagenomic sequencing demonstrated a significantly higher rate of positive microbiological findings as compared to conventional microbiology tests (CMT) (334/404 (82.7%) vs. 159/404 (39.4%), $P < 0.001$). For the 343 infectious episodes, the sites of infection among the three groups were shown in Figure 1. We observed differences in microbial detection rates of both CMT and mNGS among the three groups. The highest detection rate was seen in group B, followed by group C and A, as shown in the mNGS testing (A-68.7%, B-96.1%, C-78.8%, $P < 0.001$) and CMT (A-24.1%, B-53.9%, C-42.6%, $P < 0.001$), respectively. Our results indicated that *Klebsiella*, *Escherichia*, and *Pseudomonas* spp. were the most-frequently isolated bacterial species after allo-HSCT, which were more likely to be present in patients of pre-engraftment group than others.

We compared the detection rates of bacteria and fungi in infectious episodes with and without GVHD. The detection rate of bacteria was significantly higher in GVHD cases, irrespective of the diagnostic method used (mNGS, GVHD vs no-GVHD 37.3% vs. 22.2%, $P = 0.008$; CMT, GVHD vs no-GVHD 20.6% vs. 9.5%, $P = 0.011$). The highest detection rate was observed in cases with Grade III-IV acute GVHD. Similarly, the same trend was seen in fungal detection, although the differences were not statistically significant. Nonetheless, when only *Aspergillus* spp. and *Mucorales* were considered, cases with GVHD exhibited higher detection rates than those without GVHD (*Aspergillus*: 11.8% vs. 5.1%, $P = 0.048$; *Mucorales*: 5.9% vs. 1.3%, $P = 0.035$). Among the 404 mNGS tests, 377 of which underwent microbiological culture within 48 hours of sample collection, and the positive rate of culture was 10.1% (38/377). Using culture results as reference, the sensitivity and specificity of mNGS were 97.4% and 64.4%, respectively. Using all relevant tests as reference (CMT), the sensitivity and specificity of mNGS were 86.7% and 26.8%, respectively (Table 1).

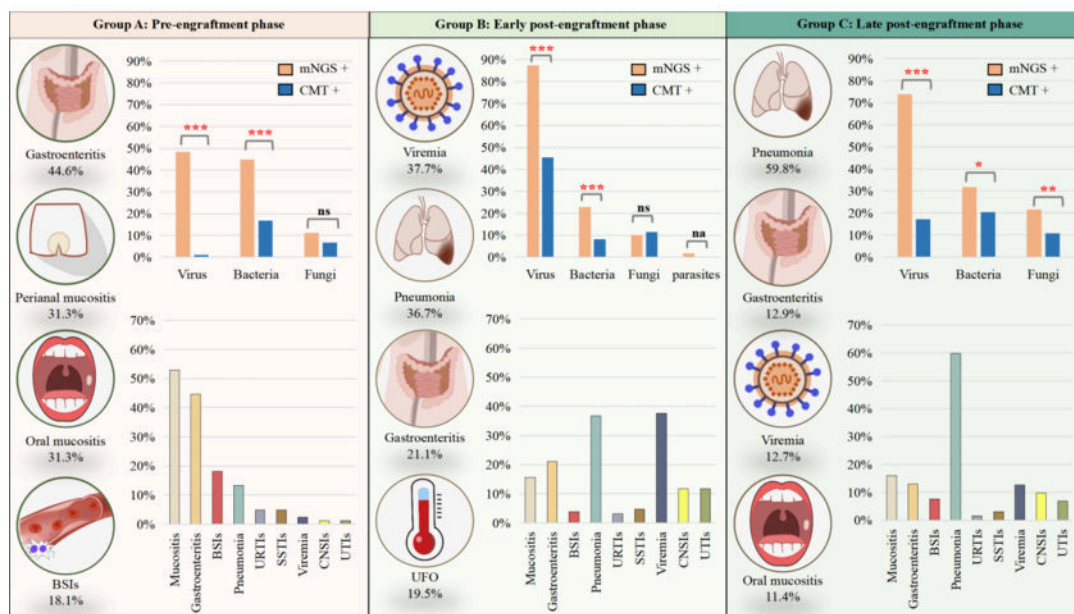
A total of 44.8% (181/404) of mNGS tests had a positive impact on etiological diagnosis (Table S1), and 24.3% (98/404) led to proper antibiotic adjustment. Further analysis was conducted on cases having positive or negative CMT results. In the positive CMT group, 21.0% (85/404) of mNGS tests had a positive impact, primarily because it detected more microorganisms that were considered pathogens by clinicians. In the negative CMT group, 23.8% (96/404) of mNGS test showed a positive impact, as it provided etiological clues. In 20.3% (82/404) of cases, the microorganisms reported by mNGS were not considered pathogenic by clinical assessment. In terms of antibiotic adjustment, mNGS tests yielded positive effects in 11.9% (48/404) and 12.4% (50/404) of cases in the positive and negative CMT groups, respectively.

The positive impact of mNGS was more evident in non-plasma samples, such as BALF and cerebrospinal fluid (Figure 3 D). For pathogen diagnosis, the positive impact was 62.7% in non-plasma samples versus 41.2% in plasma samples ($P = 0.001$). For antibiotic adjustment, the positive impact was 40.3% in non-plasma samples compared to 21.1% in plasma samples ($P = 0.002$). Furthermore, mNGS showed greater benefits in etiological diagnosis for patients with multi-site infections compared to those with single-site infections or fever of unknown origin (61.6% vs. 42.5% and 19.2%, respectively, $P = 0.001$).

Conclusion: Metagenomic sequencing had great potential for pathogen diagnosis in patients undergoing allo-HSCT. With improved methodologies and cost reduction, mNGS can serve as a valuable tool in addition to CMT.

Disclosures No relevant conflicts of interest to declare.

Figure 1. Distribution of the infection sites involved and the comparison of the detection rates among three phases after allo-HSCT.



Abbreviations: mNGS, metagenomic next-generation sequencing; CMT, conventional microbiological tests; BSIs, blood stream infections; SSTIs, skin and soft-tissue infections; UTIs, urinary tract infections; CNSIs, central nervous system infections; URTI, upper respiratory tract infections; * $P < 0.05$; ** $P < 0.01$; *** $P < 0.001$; ns, no statistically significant; na, not appropriate.

Table 1. The diagnosis performance of mNGS results versus of culture, culture and PCR, and conventional microbiology tests.

	mNGS Positive	mNGS Negative	Sensitivity(%)	Specificity(%)	PPV(%)	NPV(%)
Culture positive	37	1	97.4	64.4	23.3	99.5
Culture negative	122	217				
Culture or PCR positive	100	2	98.0	27.7	31.8	97.6
Culture andPCR negative	214	82				
CMT positive	137	21	86.7	26.8	43.2	75.9
CMT negative	180	66				

Abbreviations: mNGS, metagenomic next generation sequencing; PPV, positive predictive value; NPV, negative predictive value; PCR, polymerase chain reaction; CMT, conventional microbiology tests

Figure 1

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